

Disentangling Representation Evolution in Transformers through Directional Decomposition

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Abstract

Transformer layers evolve representations through learned additive transformations that combine parallel scaling and perpendicular steering. Across pretrained models, we find that learned updates consistently contain substantial parallel components. To test whether these components are redundant or behaviorally important, we decompose updates in two complementary spaces: residual space, analyzing complete sub-layer updates relative to the incoming hidden state; and attention value space, analyzing pre-output-projection aggregation relative to the token’s own value. Targeted edits reveal a consistent directional and space-dependent asymmetry: perpendicular edits are consistently disruptive, whereas parallel edits are comparatively redundant, especially in value space when preserving the direct self message while scaling cross-token aggregation. The same geometry informs compression and training: better-performing compressed models show lower perpendicular transformation error, and suppressing parallel attention components during from-scratch pretraining improves downstream averages in our evaluated runs, with the value-space variant strongest. Overall, this geometry connects update direction to editing robustness, compression diagnostics, and training-time intervention. Code is available in the [project repository](#).

1 Introduction

Transformer layers evolve representations through learned additive updates (Vaswani et al., 2017; Elhage et al., 2021) on top of residual streams (He et al., 2016). Geometrically, each update performs two distinct roles relative to the incoming representation: a *parallel component* that modulates magnitude while preserving direction, and a *perpendicular component* that redirects it into new

semantic subspaces. This dichotomy reframes representation evolution as a problem of *functional geometry*: decomposing learned updates by their directional role and testing their behavioral sensitivity through targeted interventions.

Across diverse pretrained models, our measurements reveal that learned updates consistently contain substantial components parallel to their incoming states. Geometrically, parallel updates amount to simple scalar rescaling, which the residual stream already provides at zero parameter cost (He et al., 2016; Elhage et al., 2021). Yet, heavily parameterized attention and MLP sub-layers actively allocate capacity to produce such components. This paradox prompts a fundamental question: are these parallel updates behaviorally essential to Transformer capabilities, or are they functionally redundant?

To resolve this question, we formulate directional decomposition across two complementary representation spaces: *residual space*, which analyzes sub-layer updates relative to the incoming hidden state, and *attention value space*, which analyzes internal value aggregation relative to the token’s own value representation. Using component-scaling interventions, we systematically modulate the parallel and perpendicular components in these spaces to measure their behavioral sensitivity.

Our interventions reveal a pronounced directional asymmetry across both representation spaces. Perpendicular scaling is consistently disruptive, confirming that orthogonal steering is essential to model capabilities. In contrast, parallel scaling is comparatively benign, leaving performance near baseline across a wide scaling interval. Within this parallel resilience, the degree of stability depends on the representation space: in attention value space, scaling the parallel component of cross-token aggregation keeps model behavior re-

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markably close to baseline, whereas residual-space parallel scaling is comparatively less stable. Thus, parallel updates exhibit broad resilience compared to perpendicular steering, especially within internal value aggregation.

Finally, this directional geometry extends beyond inference-time editing to post-training compression and from-scratch pretraining. For model compression (Frantar et al., 2023; Lin et al., 2024; Sun et al., 2024; Men et al., 2024; He et al., 2026), decomposing update distortion reveals that better-performing quantized and pruned models consistently exhibit lower perpendicular error, whereas parallel error is far less discriminative. Motivated by the inference-time asymmetry, we separately evaluate parallel attention suppression during pre-training. Across model scales from 300M to 2.7B parameters, the value-space variant consistently yields lower validation-loss trajectories, and both variants improve average downstream benchmark scores, with the value-space variant producing the largest gains.

In summary, the contributions of this work are as follows:

1. This work reveals substantial direction-preserving components in learned transformer updates and maps their behavioral robustness across representation spaces.
2. For attention updates, value-space decomposition separates direct self-value flow from cross-token aggregation, revealing a marked stability gap between value-space and residual-space edits.
3. For compression and pretraining, this directional geometry extends beyond editing: perpendicular error tracks compressed-model quality, while parallel suppression consistently improves validation loss and downstream performance across model scales.

2 Related Work

Residual-stream geometry and residual scaling. Residual connections preserve the input representation while allowing each block to add a learned transformation (He et al., 2016). In transformers, this creates a residual stream that carries information across layers and serves as the workspace for attention and feed-forward updates (Elhage et al., 2021). This preservation path is also a practical target for architectural scaling mechanisms: ReZero (Bachlechner et al., 2021), Layer-

Scale (Touvron et al., 2021), and deep-transformer initialization or normalization schemes (Xiong et al., 2020; Wang et al., 2024) adjust residual branch scale to improve optimization and stability. Residual paths also mitigate degeneracy in deep self-attention, including rank collapse (Dong et al., 2021). These works establish the importance of residual pathways and residual magnitude. We study a complementary geometric property of the learned update itself: whether it preserves the current direction or rotates the representation into a perpendicular component.

Context modeling and attention geometry. Attention explicitly aggregates information across tokens (Vaswani et al., 2017), but attention mass also concentrates on special or persistent positions, as shown by attention-sink behavior in long-context inference (Xiao et al., 2024). Gated attention variants mitigate such sink behavior by modulating attention outputs (Qiu et al., 2025). However, even when attention sinks are reduced, large diagonal attention weights still route substantial mass back to the current token itself. Exclusive Self-Attention (XSA), introduced by Zhai (2026), directly addresses this self-directed pathway by modifying value flow aligned with the current token. We revisit this issue from a parallel/perpendicular geometric perspective, focusing on when attention updates preserve the current direction and when they introduce direction-changing information.

3 Background

3.1 Parallel and Perpendicular Decomposition

Parallel and perpendicular decomposition separates a vector into components along and orthogonal to a reference direction in the same representation space. Let $\Delta \in \mathbb{R}^d$ denote the vector being analyzed and let $\mathbf{z} \in \mathbb{R}^d$ be the reference vector. The unique components are:

$$\Delta_{\parallel} = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2} \mathbf{z} = \alpha \mathbf{z}, \quad \alpha = \frac{\Delta \cdot \mathbf{z}}{\|\mathbf{z}\|^2}, \quad (1)$$

$$\Delta_{\perp} = \Delta - \Delta_{\parallel}, \quad \Delta_{\perp} \perp \mathbf{z}. \quad (2)$$

The parallel component $\Delta_{\parallel}$ lies along the reference direction $\mathbf{z}$, whereas $\Delta_{\perp}$ lies in its $(d - 1)$ -dimensional orthogonal subspace and carries the direction-changing degrees of freedom.

This is the basic decomposition used throughout the paper. In residual space, Δ is a contribution added to the hidden state; in value space, it is the value aggregate formed inside attention.

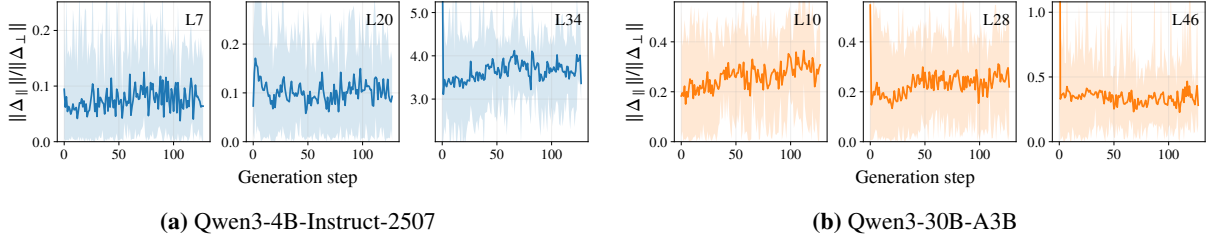


Figure 1: Parallel components persist across depth. For each accumulated attention-plus-MLP update, $\Delta_{t,\parallel}^l$ is its projection along the incoming hidden state, whereas $\Delta_{t,\perp}^l$ is orthogonal to that state. Curves show $\|\Delta_{t,\parallel}^l\|/\|\Delta_{t,\perp}^l\|$ at three sampled layers per model across generation steps; lines are means and bands span the observed range. Ratios above one indicate parallel dominance.

3.2 Transformer Residual Updates

We consider transformer residual streams with L layers and hidden dimension d . Each residual sub-layer adds an update to the current representation:

$$\mathbf{z}^{l+1} = \mathbf{z}^l + \Delta^l. \quad (3)$$

The update Δ^l denotes the output of a multi-head self-attention (Attn) sub-layer, a feed-forward (FFN) sub-layer, or the accumulated attention-plus-MLP contribution. For an individual sub-layer, the reference is the hidden state immediately before its residual addition. For the joint view, the reference is the state entering attention and Δ^l is the net change after both attention and MLP. We write $\mathbf{z}_t^l$ for the reference state at token position $t \in \{1, \dots, T\}$ and Δ_t^l for the corresponding residual contribution.

For a residual update:

$$\Delta_{t,\parallel}^l = \alpha_t^l \mathbf{z}_t^l, \quad \alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}. \quad (4)$$

Adding the parallel component therefore rescales the existing hidden state by $1 + \alpha_t^l$. The resulting representation is

$$\mathbf{z}_t^{l+1} = (1 + \alpha_t^l) \mathbf{z}_t^l + \Delta_{t,\perp}^l, \quad \Delta_{t,\perp}^l \perp \mathbf{z}_t^l. \quad (5)$$

To measure update geometry, we report the relative magnitude $\|\Delta_{t,\parallel}^l\|/\|\Delta_{t,\perp}^l\|$ across layers. This ratio quantifies how much update magnitude lies along the current representation versus orthogonal directions; the sign of α_t^l indicates whether the parallel component points with or against the current representation.

Figure 1 reports this ratio across generation steps at three sampled layers for two Qwen3 models. The persistent parallel components motivate us to ask when they affect model behavior and in which representation spaces those effects are robust.

4 Method

We instantiate the decomposition from §3.1 in two complementary spaces: the residual stream, where it characterizes learned updates relative to the evolving hidden state, and attention value space, where it characterizes token aggregation before output projection. These views yield the component-scaling interventions and compression diagnostics used throughout the paper. Figure 2 summarizes the intervention sites and their associated diagonal view.

4.1 Residual-Level Decomposition

Instantiation. Set $\Delta = \Delta_t^l$ (the residual update of block f^l at token t) and $\mathbf{z} = \mathbf{z}_t^l$ (the pre-update hidden state). For layer and module profiles, we report calibration-corpus averages of the per-token residual parallel ratio:

$$r_t^l = r(\Delta_t^l, \mathbf{z}_t^l) = \frac{|\alpha_t^l| \|\mathbf{z}_t^l\|}{\|\Delta_{t,\perp}^l\|}, \quad (6)$$

$$\alpha_t^l = \frac{\Delta_t^l \cdot \mathbf{z}_t^l}{\|\mathbf{z}_t^l\|^2}.$$

This residual-level view serves as the shared observable for the rest of the paper. The resulting diagnostic is block-agnostic. We apply it to (i) the Attn sub-layer output, (ii) the FFN sub-layer output, and (iii) their joint effect at the transformer-block level.

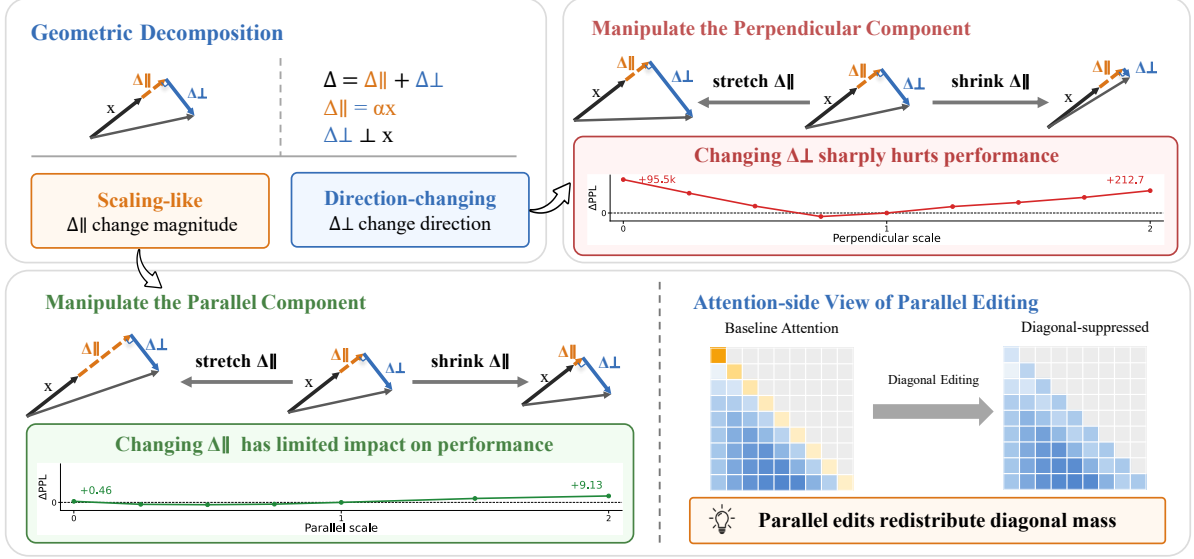


Figure 2: Geometric decomposition and component scaling. Residual updates and attention value aggregates are decomposed into parallel and perpendicular components and scaled independently; parallel-only scaling can be expressed as an attention-diagonal change. Labels report raw ΔPPL relative to the scale-one no-op (lower is better); curves are rescaled only for display.

4.2 Value-Space Decomposition

Inside attention, let $\mathbf{x}_s^{\text{attn}}$ denote the input to the value projection, including the model’s pre-attention normalization. Each source token s is mapped to a value vector $\mathbf{v}_s = W_V \mathbf{x}_s^{\text{attn}}$. For query token t , let $\mathcal{A}_{ts}$ denote a linear attention-mixing operator on the value space. This notation does not assume a particular head structure; standard multi-head attention is the block-diagonal instance whose head blocks carry their corresponding attention weights. The pre-output-projection aggregate is

$$\mathbf{o}_t = \sum_{s \leq t} \mathcal{A}_{ts} \mathbf{v}_s. \quad (7)$$

Within this space the natural reference is the current token’s own value $\mathbf{v}_t$: the direction $\mathbf{o}_t$ would take if attention routed nothing from other positions.

$$\mathbf{o}_{t,\parallel} = \frac{\mathbf{o}_t^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{o}_{t,\perp} = \mathbf{o}_t - \mathbf{o}_{t,\parallel}. \quad (8)$$

Residual-space geometry does not isolate this self-value-aligned structure because the output projection mixes the aggregate into the residual stream.

Direct Self-Message Control. Attention value aggregation inherently combines two distinct computational pathways: identity preservation via the token’s own value ($\mathcal{A}_{tt} \mathbf{v}_t$) and contextual routing from prior positions ($\sum_{s < t} \mathcal{A}_{ts} \mathbf{v}_s$). Perturbing

both simultaneously conflates cross-token aggregation with the trivial erasure of the query token’s identity. As a rigorous measurement control, we decouple these pathways and apply directional decomposition exclusively to the non-self aggregate:

$$\mathbf{o}_t = \underbrace{\mathcal{A}_{tt} \mathbf{v}_t}_{\mathbf{d}_t \text{ (self)}} + \underbrace{\sum_{s < t} \mathcal{A}_{ts} \mathbf{v}_s}_{\mathbf{c}_t \text{ (non-self)}}, \quad (9)$$

$$\mathbf{c}_{t,\parallel} = \frac{\mathbf{c}_t^\top \mathbf{v}_t}{\|\mathbf{v}_t\|^2} \mathbf{v}_t, \quad \mathbf{c}_{t,\perp} = \mathbf{c}_t - \mathbf{c}_{t,\parallel}.$$

We scale the directional components of the non-self aggregate while preserving the untouched self message:

$$\hat{\mathbf{o}}_t^{\text{excl}} = \mathbf{d}_t + s^{(\parallel)} \mathbf{c}_{t,\parallel} + s^{(\perp)} \mathbf{c}_{t,\perp}. \quad (10)$$

We designate this control *exclude-self* scaling: it isolates the directional effects of cross-token aggregation while keeping the identity carrier and attention row normalization intact.

Value-space decomposition applies only to attention aggregates. For MLPs, the main analysis decomposes the sub-layer output in residual space; Appendix D separately applies the same projection geometry to the post-gating carrier and grouped down-projected contributions.

4.3 Component Scaling and Diagonal Form

For either attention-side site, write $\mathbf{y}_t = \sum_{s \leq t} \mathcal{M}_{ts} \mathbf{m}_s$ and decompose $\mathbf{y}_t$ relative to

its reference $\mathbf{r}_t$. Value-space editing uses $(\mathcal{M}_{ts}, \mathbf{m}_s, \mathbf{r}_t) = (\mathcal{A}_{ts}, \mathbf{v}_s, \mathbf{v}_t)$. Residual-space attention editing uses $\mathcal{B}_{ts} = W_O \mathcal{A}_{ts}$ and $(\mathcal{M}_{ts}, \mathbf{m}_s, \mathbf{r}_t) = (\mathcal{B}_{ts}, \mathbf{v}_s, \mathbf{z}_t)$, keeping head-dependent mixing explicit. Component scaling then uses the shared intervention

$$\tilde{\mathbf{y}}_t = s^{(\parallel)} \mathbf{y}_{t,\parallel} + s^{(\perp)} \mathbf{y}_{t,\perp}. \quad (11)$$

The no-op is $s^{(\parallel)} = s^{(\perp)} = 1$, and parallel removal sets $s^{(\parallel)} = 0, s^{(\perp)} = 1$. Exclude-self scaling applies the same form to $\mathbf{c}_t$ and restores $\mathbf{d}_t$ as in Eq. 10.

The same edit can also be read on the attention map itself, connecting it to self-directed attention mass (§2). Holding all off-diagonal messages fixed, a diagonal update realizes parallel-only scaling when

$$\Delta \mathbf{y}_{tt} = (s^{(\parallel)} - 1) \mathbf{y}_{t,\parallel}. \quad (12)$$

Appendix A gives an operator realization, the scalar attention-diagonal forms used in our visual comparison, their removed cases, and the applied-output audit.

5 Experimental Setup

Inference-time interventions. For pretrained-model evaluations, edits operate only during the forward pass and leave model weights unchanged. Value-space results use the self-preserving scaling in Eq. 10 unless explicitly labeled otherwise; each experiment reports its edited layers and scales. Appendices A–B give implementation details.

Pretraining setup. For pretraining, we train three GPT-style configurations from scratch on standard next-token prediction, with baseline and full-aggregate parallel-removal variants. We additionally train 1.4B and 2.7B models for downstream evaluation (Table 3). Model configurations and the released settings are in Appendix E.

6 Directional Robustness of Updates

6.1 Component Scaling

Component-scaling results. Figure 3 sweeps $s^{(\parallel)}$ over $[-1, 3]$ and $s^{(\perp)}$ over $[0, 2]$ at three sites. On the parallel grid, exclude-self value-space scaling is stable within the unit interval: full removal changes PPL by only 0.46 and the largest absolute change is 1.32, whereas removing either residual-space parallel component increases PPL by about

21–22. The stable interval has a clear sign boundary: under reversal, the value-space delta reaches 188.77 at scale -1 . Perpendicular scaling produces larger changes: at scale 2, the deltas are 212.68 in value space, 14,757.67 for the residual attention branch, and 52.62 for the residual MLP branch. The sweep therefore shows both directional and representation-space asymmetries. Residual-space edits act on a complete learned branch, whereas exclude-self scaling isolates the non-self parallel component while preserving the remaining aggregate. The same directional asymmetry is already present inside the MLP: decomposing the post-gating carrier rather than the branch output leaves full parallel removal near baseline while perpendicular removal remains disruptive (Appendix D).

Appendix A formalizes this distinction. *The scaling sweep uses component editing as a functional probe. We next turn the probe into a sharper removal test of whether value-space parallel removal preserves performance on standard downstream and long-context evaluations.*

6.2 Value-Space vs. Residual-Space Editing

Robustness on general tasks. Table 1 reports the full-removal comparison on the seven non-generative tasks. Under the matched residual/diagonal protocol, Attn Para-Rem. changes the average by $-2.46 / -2.94$ points on the 1.7B/30B-A3B models, whereas Diag. Rem. changes it by $-22.44 / -32.97$ points. Under the value-space protocol, exclude-self removal stays within 1.51 points of the 1.7B baseline and 0.12 points of the 30B-A3B baseline, whereas editing the self contribution as well reduces the average by 8.2–10.0 points. Exclude-self removal is markedly more robust because it preserves the self edge and leaves the remaining attention normalization unchanged; hard diagonal removal is far more disruptive because it removes the self edge and renormalizes the remaining attention. Thus robustness depends on both the representation space and preservation of the direct self message.

To test whether self-preserving value-space scaling remains effective as the available context grows, Table 2 extends the comparison from 4k to 12k. Across all three context lengths, retaining half remains comparatively stable with the self term fixed, and the exclude-self variant consistently outperforms full-aggregate scaling at the same retained scale. At 4k, the retained-scale results on Gemma-3-12B-IT and Llama-3.2-3B remain comparable to

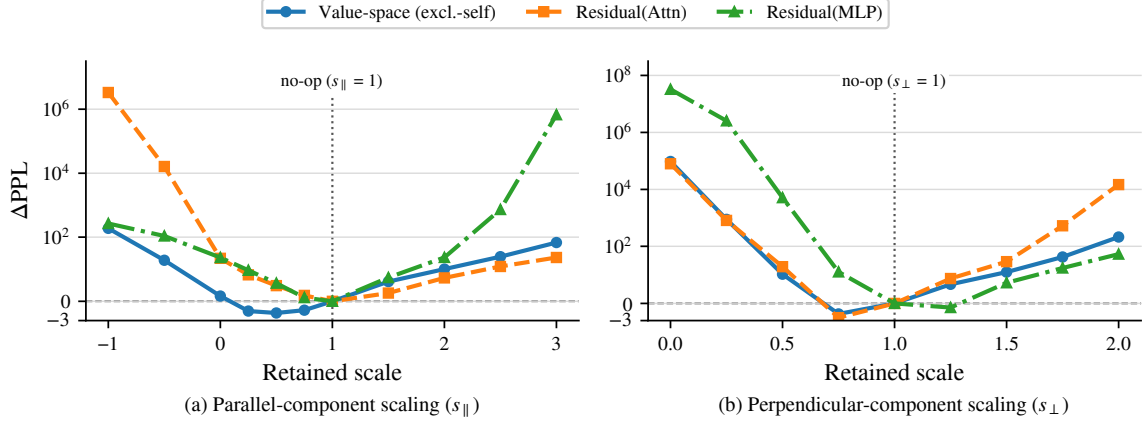


Figure 3: Perplexity under component scaling on Qwen3-1.7B. The left panel varies $s_{\parallel}$ with $s_{\perp} = 1$; the right panel varies $s_{\perp}$ with $s_{\parallel} = 1$. On WikiText-2, ΔPPL is edited minus the matched scale-one no-op, so lower is better. All decoder layers are edited; value-space curves use exclude-self scaling.

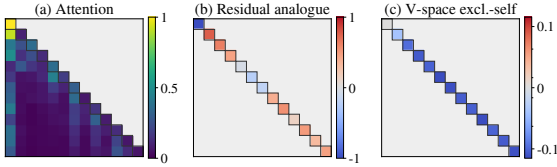


Figure 4: Effective diagonal views on Qwen3-4B at layer 19. For full parallel removal ($s_{\parallel} = 0$), the left panel shows head-6 attention weights; the middle shows the residual-space attention-map view, and the right shows the pre- W_O , head-6 exclude-self value-space change. The latter two color scales encode ΔA_{tt} ; off-diagonal entries are unchanged.

their matched baselines (Appendix Table 8).

Having established the behavioral gap between value- and residual-space edits, we next inspect how their parallel-removal objectives appear as diagonal coefficient changes.

Attention diagonal views for parallel editing.

Figure 4 compares the value-space diagonal change with the residual-space attention-map view; Appendix Table 4 lists both coefficients.

The residual-space coefficients span -1.00 to 0.76 , whereas the selected head’s exclude-self value-space coefficients are non-positive, with the first-token coefficient exactly zero because there is no non-self context. The difference reflects their reference directions and the post- W_O mixing used by residual-space editing. Coefficient magnitude is not intervention magnitude: a Qwen3-0.6B diagnostic measures both edits at the same post- W_O attention output and finds perturbation norms of 25.15% of the original branch norm for the residual-space edit, versus 49.10% for exclude-self value-

space editing (Appendix Table 5). Residual-space fragility therefore does not follow from a larger applied perturbation in this comparison.

The all-layer averages also conceal substantial head and depth heterogeneity. Appendix C reports isolated-head sensitivities and joint all-head controls across four Qwen3 scales.

6.3 Compression Diagnostics

The removal experiments show that model behavior is especially sensitive to direction-changing components of activation updates. We next ask how this picture appears when the model itself is modified. Compression changes the parameters, so its effect is not an explicit activation edit; instead, it induces an error between the baseline and compressed-model updates. We decompose this error to test whether its perpendicular component distinguishes compression settings more clearly than its parallel component.

At each layer, we define the compression error as $e = \Delta_{\text{comp}} - \Delta_{\text{base}}$ and decompose it relative to the baseline-update direction: $e_{\parallel} = \text{proj}_{\Delta_{\text{base}}} e$ and $e_{\perp} = e - e_{\parallel}$. Figure 6 reports the component norms normalized by $\|\Delta_{\text{base}}\|$ and averaged over fixed prompts.

Perpendicular-error curves separate the four settings more clearly, while parallel-error curves interleave. Thus the direction-changing error provides the more discriminative component-level diagnostic in this comparison. Appendix F provides branch-wise views, the derivation, and the supporting baseline-update magnitude plot.

Table 1: Task-level comparison of residual-, value-space, and diagonal controls. Entries are absolute scores (%); higher is better and Avg. is unweighted. Both value-space rows set $s_{||} = 0$. V-Para Rem. edits the full value aggregate; V-Excl.-self preserves $A_{tt}v_t$. Diag. Rem. zeros A_{tt} and renormalizes over non-self positions for $t > 0$; no edit is applied at $t = 0$. Attn Para-Rem. and Diag. Rem. use separate matched baselines. Differences reported in the text are computed before rounding. Bold marks the best baseline/value-space result.

Model	Setting	OBQA	PIQA	Wino	ARC-C	HSwag	BoolQ	RTE	Avg.
Qwen3-1.7B	Baseline	37.20	72.47	60.69	53.07	60.16	77.65	70.04	61.61
	Attn Para-Rem.	38.20	72.31	56.59	50.09	58.74	71.38	66.79	<u>59.16</u>
	V-Para Rem.	34.40	69.48	54.85	41.38	54.04	66.27	53.43	<u>53.41</u>
	V-Excl.-self	39.20	71.76	60.54	49.15	59.88	74.46	65.70	<u>60.10</u>
	Diag. Rem.	26.20	51.36	49.17	24.49	26.66	40.70	55.60	<u>39.17</u>
Qwen3-30B-A3B	Baseline	44.80	80.69	70.24	69.97	77.79	88.53	81.95	73.42
	Attn Para-Rem.	44.00	78.73	66.85	65.10	73.87	88.04	78.34	<u>70.70</u>
	V-Para Rem.	41.20	76.22	64.56	59.04	69.57	70.89	62.45	<u>63.42</u>
	V-Excl.-self	44.60	79.87	72.45	68.34	78.13	86.36	83.39	<u>73.31</u>
	Diag. Rem.	30.60	57.67	48.46	26.96	34.78	38.59	47.65	<u>40.68</u>

Table 2: RULER on Llama-3.2-3B. Scores are means over all 13 tasks; parentheses give edited minus matched baseline, computed before rounding. Scale 0 removes the parallel component and 0.5 retains half. V-Full scales the whole value aggregate; V-Excl.-self preserves the direct self term and scales only the remainder.

Method	4k	8k	12k
Baseline	86.91	82.09	79.30
V-Full, $s_{ }=0.5$	84.81 (-2.10)	78.80 (-3.29)	74.76 (-4.53)
V-Excl.-self, $s_{ }=0.5$	86.69 (-0.22)	80.80 (-1.29)	76.88 (-2.42)
V-Full, $s_{ }=0$	60.99 (-25.92)	48.61 (-33.48)	39.17 (-40.12)
V-Excl.-self, $s_{ }=0$	75.64 (-11.27)	67.97 (-14.11)	62.02 (-17.27)

7 Training-Time Allocation

The inference-time results show that the effect of suppressing parallel components depends strongly on the representation space in which the intervention is applied. We therefore ask whether applying the same intervention throughout training changes how optimization allocates learning across the retained, direction-changing updates. We train from scratch with residual- or value-space parallel removal active during both optimization and evaluation, avoiding a train-inference mismatch. We compare the two interventions through their training trajectories and downstream scores; Appendix E gives the configurations and gated-scaling control.

Figure 5 shows that suppressing parallel attention updates changes the optimization trajectory. The separation from baseline appears early and remains visible across all three configurations through the final checkpoint, rather than emerging only as an endpoint difference.

Table 3 reports downstream evaluation for the

Table 3: Post-training benchmark summary. Avg. is the unweighted mean accuracy over ARC-Easy, BoolQ, HellaSwag, OpenBookQA, PIQA, and WinoGrande; Δ Avg is relative to each model’s matched baseline. Attn and V-Para denote residual-space attention and value-space parallel removal.

Model	Setting	Avg	Δ Avg
1.4B	Baseline	58.5	0.0
	Attn Para-Rem.	58.8	+0.3
	V-Para Rem.	59.2	+0.7
2.7B	Baseline	60.2	0.0
	Attn Para-Rem.	60.9	+0.7
	V-Para Rem.	61.7	+1.5

larger configurations. Attn Para-Rem. improves the average over baseline, and V-Para Rem. gives the strongest average at both scales, with gains of 0.7 and 1.5 points. Together with the loss curves, these results show that parallel suppression affects both the training trajectory and the resulting checkpoint, with a larger downstream change for the value-space variant.

Appendix E gives task-level scores for both larger models and the gated-scaling control, where fixed removal retains the lowest training loss.

8 Discussion

Our findings establish directional decomposition not merely as an analytical probe, but as a lens that illuminates fundamental mechanisms of Transformer computation across their lifecycle.

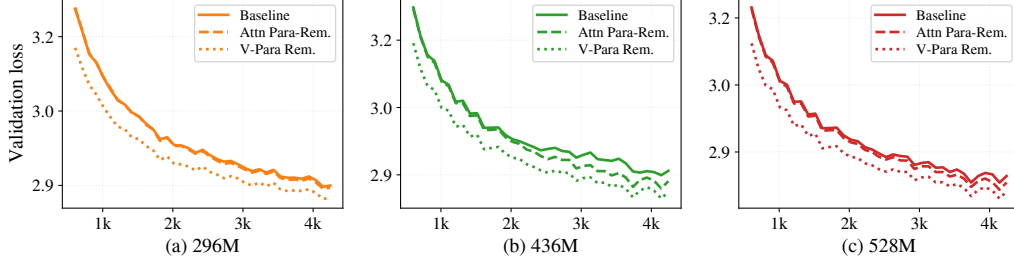


Figure 5: Training-time parallel removal lowers validation loss. The displayed OpenWebText (Gokaslan et al., 2019) loss trajectories show baseline, Attn Para-Rem., and V-Para Rem. for matched configurations. Both edits set $s_{\parallel} = 0$ and retain the perpendicular component.

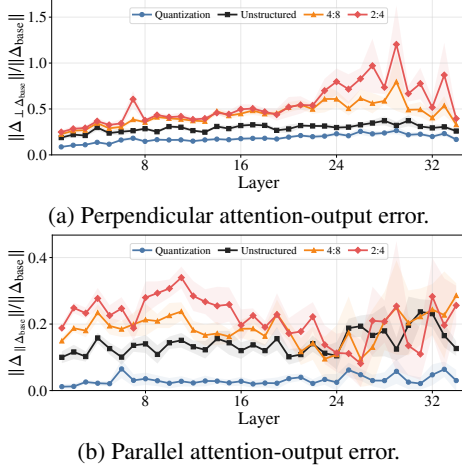


Figure 6: Directional compression error. Curves are layerwise means, with bands showing one standard deviation over 12 fixed prompts, comparing compressed and baseline Qwen3-4B attention updates. The upper and lower panels show perpendicular and parallel error, each normalized by the full baseline-update norm (higher is more distortion). Settings are 4-bit AWQ and 50% Wanda pruning with unstructured, 4:8, or 2:4 masks.

Geometric Duality of Representation Evolution. Standard formulations treat Transformer layer depth as incremental vector accumulation $\mathbf{x} + \Delta$. Our decomposition reveals that this accumulation executes two fundamentally distinct geometric functions: *magnitude modulation* ($\Delta_{\parallel}$) and *semantic steering* ($\Delta_{\perp}$). Perpendicular updates rotate representations into new orthogonal subspaces where factual associations and syntactic distinctions reside; because these semantic coordinates are finely calibrated, $\Delta_{\perp}$ is hyper-fragile to perturbation. Conversely, parallel updates act as adaptive gain controllers, amplifying existing feature trajectories without altering their directional meaning. The extensive tolerance basin of $\Delta_{\parallel}$ proves that Transformers possess an inherent resilience to scale variations along their representational trajectories, explaining why coarse semantic intent persists even

under substantial magnitude disruption.

Rethinking Model Compression: Beyond Isotropic L_2 Error. A ubiquitous convention in post-training quantization and pruning is the minimization of isotropic reconstruction error, $\min \|\Delta - \hat{\Delta}\|_2^2$. However, isotropic L_2 error implicitly assumes uniform sensitivity across all dimensions of $\mathbb{R}^d$. Our directional framework exposes the flaw in this assumption: approximation errors aligned with the update ($\Delta_{\parallel}^{\text{err}}$) fall within the model’s natural tolerance basin, whereas orthogonal deviations ($\Delta_{\perp}^{\text{err}}$) directly distort semantic trajectories. This resolves the long-standing mystery of why compression methods exhibiting comparable global L_2 error often diverge dramatically in downstream performance. Measuring compression distortion through directional decomposition establishes $\Delta_{\perp}^{\text{err}}$ as a causal, high-fidelity predictor of downstream degradation, suggesting that future compression objectives should penalize directional rotation rather than uniform distance.

Architectural Implications and Pretraining Dynamics. Why do unconstrained Transformers consistently allocate substantial parameter capacity to redundant parallel scaling within attention? In standard self-attention, token mixing is structurally entangled with self-representation amplification. Our from-scratch pretraining results demonstrate that suppressing parallel updates relieves attention from redundant scaling duties, guiding optimization to dedicate its capacity toward directional contextual steering. This architectural inductive bias accelerates convergence, lowering validation-loss trajectories and improving downstream generalization across model scales from 300M to 2.7B parameters. These findings provide principled geometric support for emerging architectures that decouple contextual routing from magnitude modulation (such as gated attention and query-key normaliza-

tion), indicating that directional orthogonality is a promising inductive bias for next-generation foundation models.

9 Conclusion

In this work, we formalized representation evolution in Transformers through directional orthogonal decomposition, separating learned updates into parallel magnitude modulation ($\Delta_{\parallel}$) and perpendicular semantic steering ($\Delta_{\perp}$). Across pretrained language models, our interventions uncovered a universal directional asymmetry: representations exhibit an expansive, resilient tolerance basin for parallel scaling, but acute, fragile sensitivity to perpendicular perturbations. We demonstrated that this geometric framework provides practical utility across the model lifecycle: resolving discrepancies in post-training compression by identifying perpendicular error as the true causal driver of degradation, and guiding from-scratch pretraining to lower validation-loss trajectories and improve downstream generalization. These results establish directional geometry as a principled foundation for deciphering internal representations, diagnosing model compression, and guiding future architecture design.

Limitations

Our experiments cover decoder-only language models across multiple scales, language tasks, inference-time interventions, compression settings, and from-scratch training configurations. Although the same geometric picture is consistent across the evaluated settings, its behavior under other architectures, modalities, and post-training regimes remains to be established. Future work should test whether these directional patterns persist across such settings and connect layer-level geometry more directly to downstream behavior. We use the proposed edits primarily as analysis tools; turning them into practical training or inference methods requires further validation.

Ethical Considerations

Directional decomposition provides a concrete tool for auditing representation change and screening compression or interventions before downstream evaluation. This screening requires no retraining and is designed to complement task-specific testing and human review. Responsible deployment should evaluate prompt- and context-specific behavior as

well as bias, privacy, security, and high-stakes use. Access controls and downstream monitoring are appropriate when deploying editing capabilities.

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A Intervention Details

Attention-diagonal editing. Equation 12 defines the requested change in the parallel component. Let $\mathbf{m}_t$ be the self message, $\mathbf{r}_t$ the reference, and $\mathcal{B}_{tt}$ its linear map into the edited space. One minimum-Frobenius-norm realization of $\Delta \mathbf{y}_{tt} = \Delta \mathcal{B}_{tt} \mathbf{m}_t$ is

$$\Delta \mathcal{B}_{tt}^* = (s^{(\parallel)} - 1) \frac{\mathbf{y}_t^\top \mathbf{r}_t}{\|\mathbf{m}_t\|^2 \|\mathbf{r}_t\|^2} \mathbf{r}_t \mathbf{m}_t^\top.$$

For a structured operator family, the same construction uses the Frobenius projection of $\mathbf{r}_t \mathbf{m}_t^\top$ onto the admissible operator subspace. Table 4 gives the resulting scalar forms. For the exclude-self variant used in our primary value-space results, Eq. 10 instead applies the projection to $\sum_{s < t} A_{ts} \mathbf{v}_s$ and then restores the unchanged direct term $A_{tt} \mathbf{v}_t$.

Table 4: Attention-diagonal forms for component scaling. The value-space form is exact; the residual form matches only the scalar projection constraint. “Removed case” sets $s^{(\parallel)} = 0$ while retaining the perpendicular component.

Quantity	Form
<i>Full-aggregate value space</i>	
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{\mathbf{v}_t^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}.$
Removed case	$\tilde{A}_{tt} = - \sum_{s < t} A_{ts} \frac{\mathbf{v}_s^\top \mathbf{v}_t}{\ \mathbf{v}_t\ ^2}.$
<i>Residual space</i>	
Diagonal change	$\Delta A_{tt} = (s^{(\parallel)} - 1) \frac{(\Delta A_{tt})^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}.$
Removed case	$\tilde{A}_{tt} = A_{tt} - \sum_{s < t} A_{ts} \frac{(W_O \mathbf{v}_s)^\top \mathbf{z}_t}{(W_O \mathbf{v}_t)^\top \mathbf{z}_t}.$

Applied strength of the two edits. Table 5 audits both Full No-Para edits at a common post- W_O site. The exclude-self value-space edit applies the larger perturbation on every measure, so its smaller behavioral cost does not follow from a gentler intervention.

Table 5: Applied strength of attention Full No-Para interventions. Full No-Para sets $s_{\parallel} = 0$. Ratios are measured at the summed attention output on Qwen3-0.6B; value-space attention preserves the self term and is mapped through W_O . Perturbation and retained norm compare $\|\tilde{y} - y\|$ and $\|\tilde{y}\|$ with $\|y\|$; energy is the sample average of $\|\tilde{y} - y\|^2 / \|y\|^2$.

Geometry	Perturbation norm	Perturbation energy	Retained norm
Attention residual-space	25.15%	9.323%	95.01%
Attention exclude-self value-space	49.10%	25.89%	83.36%

Forward-hook implementation. We apply Eq. 11 through forward hooks that replace only

the selected tensor; all subsequent computation is unchanged. Table 6 reports the measured cost.

Table 6: Measured pre- W_O projection overhead. Wall-clock increases relative to the unedited baseline are measured on one RTX 6000 Ada. BF16 inference uses batch 2: prefill has length 2,048 and decode generates one token from a 512-token static cache with CUDA Graph replay. Training uses the 0.7B model with batch 1 and length 2,048. The three inference values are ordered as Qwen3-0.6B/4B/14B.

Workload	Optimized overhead
Prefill	2.47% / 1.25% / 1.11%
Static graph decode	0.83% / 0.36% / 0.25%
Training (0.7B)	3.39%

The fused kernel combines the dot product, norm, and projection update with FlashAttention-2 (Dao, 2024).

B Evaluation Details

Evaluated models. The editing and diagnostic experiments use Qwen3 (Qwen Team, 2025), and individual figures and tables name the evaluated scale. Qwen3-4B results use the base checkpoint, except Figure 1, which uses Qwen3-4B-Instruct-2507. Long-context evaluation uses Llama-3.2-3B (Llama Team, 2024).

Benchmark evaluation. Benchmark and RULER tasks are executed through the Language Model Evaluation Harness (Gao et al., 2024). RULER (Hsieh et al., 2024) provides the long-context evaluation. Task definitions follow the original sources for CoQA (Reddy et al., 2019), DROP (Dua et al., 2019), TruthfulQA (Lin et al., 2022), and the remaining benchmarks (Clark et al., 2018, 2019; Zellers et al., 2019; Mihaylov et al., 2018; Bisk et al., 2020; Wang et al., 2018; Sakaguchi et al., 2020; Hendrycks et al., 2021; Cobbe et al., 2021). Table 7 gives the default few-shot setting and metric for each task.

Table 7: Default evaluation task settings. Acc. denotes accuracy; Norm denotes length normalization.

Task	Few-shot	Metric
ARC-Challenge	25	Acc. (Norm)
ARC-Easy	0	Acc. (Norm)
BoolQ	0	Acc.
CoQA	0	F1
DROP	0	F1
GSM8K	5	Exact match
HellaSwag	10	Acc. (Norm)
MMLU	5	Acc.
OpenBookQA	0	Acc. (Norm)
PIQA	0	Acc. (Norm)
RTE	0	Acc.
TruthfulQA-MC1/2	0	Acc.
TruthfulQA generation	0	Rouge-L
WinoGrande	5	Acc.

C Additional Editing Results

Zero-shot parallel editing. Table 8 tests whether self-preserving value-space scaling transfers across model families on RULER-4k. At $s_{||} = 0.5$, Gemma-3-12B, Llama-3.2-3B, and Qwen3-4B/8B remain within 0.5 points of their matched baselines, showing near-baseline behavior across three model families.

Table 8: Cross-model retained-scale RULER-4k evaluation. V-Excl.-self preserves the direct self contribution and sets $s_{||} = 0.5$ for the non-self aggregate; Δ is edited minus matched baseline and is computed before display rounding.

Model	Baseline	V-Excl.-self, $s_{ }=0.5$	Δ
Gemma-3-1B-IT	68.0	62.8	-5.3
Gemma-3-12B-IT	94.0	94.1	+0.1
Llama-3.2-3B	86.9	86.7	-0.2
Qwen3-0.6B	79.8	75.1	-4.8
Qwen3-1.7B	88.1	84.9	-3.2
Qwen3-4B	92.6	92.4	-0.2
Qwen3-8B	94.2	93.7	-0.5

Head-level localization. We test localization across every decoder layer of four Qwen3 scales by comparing all isolated query-head edits with a joint all-head edit. Figure 7 includes every head without significance-based selection.

Layer 0 gives the largest positive joint effect at every scale. Beyond layer 0, the joint effect is strongly depth-specific and often extends beyond the corresponding single-head range.

D MLP-Internal Geometry

Residual-space editing observes only the final MLP output, so it does not show whether the same directional asymmetry is already present before the down projection. Inside a gated MLP, the post-

gating carrier and down-projected output are

$$\begin{aligned}
 h &= \phi(W_{\text{gate}}x) \odot W_{\text{up}}x, \\
 y &= \sum_g y_g, \\
 y_g &= W_{\text{down}}[:, I_g]h_g,
 \end{aligned} \tag{13}$$

where $h_g = h[I_g]$. We choose residual-width groups, so h_g , y_g , and x have the same dimensionality. This exposes two internal editing sites: each carrier h_g is decomposed relative to x before the down projection, and each projected contribution y_g is decomposed relative to its carrier h_g before the group contributions are summed. These controls distinguish geometry already present in the gated intermediate representation from geometry introduced by the down projection. Table 9 reports fixed-token loss on Qwen3-0.6B. At the two internal sites, full parallel removal changes loss by at most 0.0301, whereas perpendicular removal increases it by more than 13 points.

Table 9: MLP-internal component removal. Values are edited-minus-no-op loss on fixed C4 tokens (lower is better). Full No-Para or No-Perp sets the named retained scale to zero; every no-op matches baseline.

Geometry	Full No-Para	Full No-Perp
Residual MLP (y relative to x)	+0.1165	+15.9567
Grouped h relative to x	+0.0301	+13.4184
Grouped y_g relative to h_g	+0.0037	+13.3324

Table 10 reports matched downstream controls beyond fixed-token loss.

E Pretraining Configuration Details

Table 11 resolves the summary averages into task-level scores. The trajectories in Figure 5 use OpenWebText, and the 1.4B/2.7B runs use FineWeb100BT from FineWeb (Penedo et al., 2024). In every setting, the full-aggregate value-space intervention is applied at every layer.

The OpenWebText configurations labeled 296M, 436M, and 528M use (layers, heads, width) of (14, 16, 1024), (16, 18, 1152), and (18, 20, 1280); the 1.4B and 2.7B configurations use (24, 16, 2048) and (32, 20, 2560). The larger runs use eight DDP processes, approximately 104.9B tokens, 2,048-token contexts, 200K updates, and global batch 256. For the 1.4B model, the per-process microbatch is 2, the accumulation factor is 128, and the peak/minimum learning rates are $4 \times 10^{-4}/4 \times 10^{-5}$; the corresponding settings for the 2.7B model are 1, 256, and $3 \times 10^{-4}/3 \times 10^{-5}$. Shared

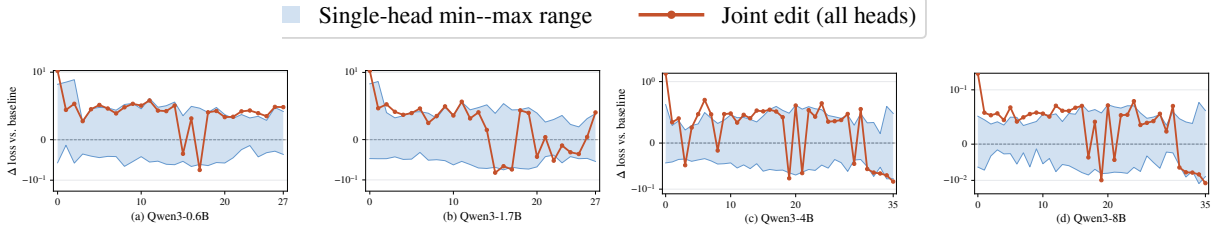


Figure 7: All-layer head sensitivity across Qwen3 scales. At each layer, orange shows joint full removal from all heads and the blue band spans the minimum and maximum isolated single-head effects; it is not a confidence interval. Δ loss is edited minus baseline over 128 fixed C4 texts (positive is worse), shown on symmetric-log axes.

Table 10: Downstream controls for MLP-internal geometry. Core-MC averages nine 0-shot benchmarks; MMLU is 5-shot. GSM8K-S/F are strict/flexible exact match, TQA R-L is TruthfulQA-generation Rouge-L, and DROP/CoQA use F1.

(a) Multiple-choice controls				
Model	Setting	Core-MC	MMLU	
Qwen3-0.6B	Baseline	50.94	52.40	
	Full No-Para	50.98	52.66	
	Full No-Perp	36.62	26.89	
Qwen3-1.7B	Baseline	55.19	60.21	
	Full No-Para	55.28	60.39	
	Full No-Perp	37.35	24.40	

(b) Generation and question-answering controls						
Model	Setting	GSM8K-S	GSM8K-F	TQA R-L	DROP	CoQA
Qwen3-0.6B	Baseline	49.43	50.42	18.98	7.76	74.24
	Full No-Para	49.96	50.72	15.61	7.66	74.04
Qwen3-1.7B	Baseline	68.76	68.92	48.34	7.32	77.06
	Full No-Para	68.08	68.01	48.32	7.44	77.69

settings are a 2K-step warm-up with cosine decay, AdamW $\beta = (0.9, 0.95)$, weight decay 0.1, gradient clipping at 1.0, no dropout, seed 1337, and 200-batch evaluation every 1K updates.

Gated parallel-scaling diagnostic. Figure 8 compares baseline, fixed V-Para Rem., and gated parallel-scaling in the 1.4B run. Fixed removal has the lowest training-loss trajectory among the three settings.

F Extended Compression Geometry

We use the normalization from Figure 6. Figure 9 extends the breakdown to the accumulated update and the MLP output, and Table 12 reports the layerwise correlation between total and perpendicular attention-output error.

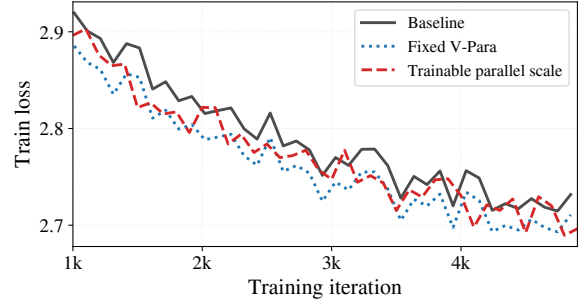


Figure 8: Parallel-scale adjustment during 1.4B pre-training. FineWeb100BT training loss (lower is better) is plotted against update. Fixed V-Para sets the full-aggregate $s_{\parallel} = 0$; trainable scaling instead predicts a tokenwise parallel scale.

Table 12: Attention-output total error is mostly aligned with perpendicular error. On Qwen3-4B and 12 fixed prompts, r and ρ are layerwise Pearson and Spearman correlations between normalized total error $\|e\|/\|\Delta_{\text{base}}\|$ and its perpendicular or parallel component. Perp./Total and Para./Total are mean component-to-total ratios across layers.

Setting	$r(e, e_{\perp})$	$\rho(e, e_{\perp})$	$\rho(e, e_{\parallel})$	Perp./Total	Para./Total
Quantization	0.998	0.990	0.596	0.980	0.168
Unstructured	0.970	0.909	0.810	0.891	0.439
4:8	0.986	0.986	-0.135	0.887	0.415
2:4	0.991	0.979	-0.432	0.875	0.425

Connecting Layer Dropping and Representation Evolution. The parallel/perpendicular view also decomposes a cosine-based diagnostic used for layer dropping (He et al., 2026). Let x be the incoming representation and let a layer update decompose as $\Delta = \Delta_{\parallel} + \Delta_{\perp}$, with $\Delta_{\parallel} = \alpha x$ and $\|\Delta_{\perp}\| = \beta\|x\|$. Then

$$\cos(x, x + \Delta) = \frac{1 + \alpha}{\sqrt{(1 + \alpha)^2 + \beta^2}}.$$

The diagnostic depends on $|\beta/(1 + \alpha)|$. In the positive-scale, small-angle regime, $1 - \cos(x, x + \Delta) \approx \frac{1}{2}(\beta/(1 + \alpha))^2$. Figure 10 reports the baseline update magnitudes.

Table 11: Task-wise post-training benchmark scores. Avg. is unweighted and Δ Avg uses the same-scale baseline. Base, Attn, and V-Para denote baseline, residual-space attention parallel removal, and full-aggregate value-space parallel removal.

Model	Setting	ARC-E	BoolQ	HSwag	OBQA	PIQA	WinoGr	Avg	Δ Avg
1.4B	Base	56.0	65.1	60.5	34.7	75.8	58.7	58.5	0.0
	Attn	57.1	63.9	61.3	35.3	76.0	59.1	58.8	+0.3
	V-Para	58.3	62.8	62.1	35.9	76.3	59.8	59.2	+0.7
2.7B	Base	58.4	61.2	66.0	37.1	76.5	61.8	60.2	0.0
	Attn	59.3	62.8	66.7	37.7	77.0	62.1	60.9	+0.7
	V-Para	60.4	64.4	67.1	38.2	77.6	62.6	61.7	+1.5

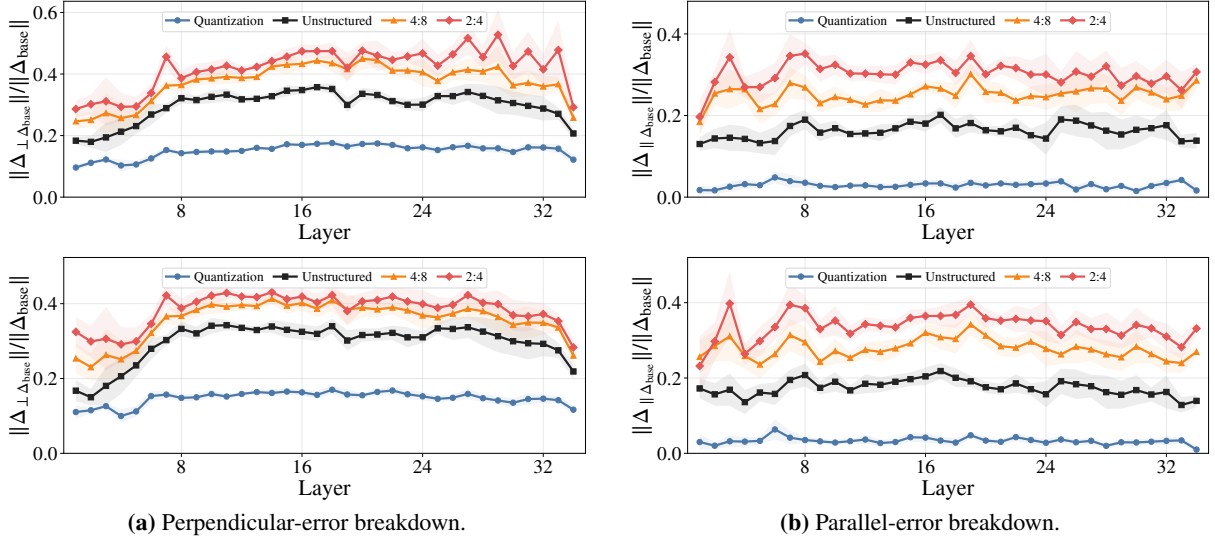


Figure 9: Compression-geometry breakdown across branches. The left and right groups show perpendicular and parallel error, respectively. Within each group, the upper and lower panels show compressed-minus-baseline error for the accumulated attention-plus-MLP update and the MLP output, each normalized by the full baseline-update norm. Curves are layerwise means, with bands showing one standard deviation over 12 fixed prompts on Qwen3-4B. Settings are 4-bit AWQ and 50% Wanda pruning with unstructured, 4:8, or 2:4 masks.

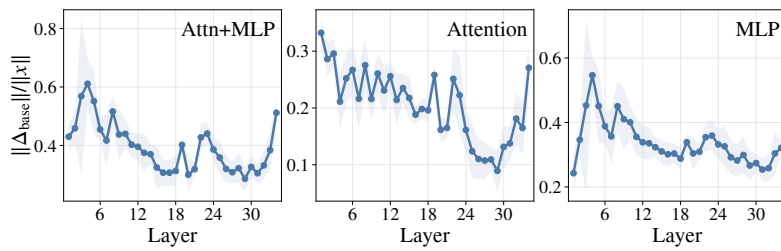


Figure 10: Baseline layer updates are small relative to the residual stream. Means and one-standard-deviation bands for $\|\Delta_{\text{base}}\|/\|x\|$ across 12 prompts for Qwen3-4B layers 1–34, shown for the accumulated attention-plus-MLP update, attention output, and MLP output. Values below one indicate updates smaller than the incoming residual stream.